

Occupational exposure to meat and risk of lymphoma: a multicenter case-control study from Europe.

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Several studies have suggested an increased risk of lymphoma among workers exposed to meat, without conclusive evidence. We conducted a multicenter case-control study during 1998-2004 in the Czech Republic, France, Germany, Ireland, Italy and Spain, including 2,007 cases of non-Hodgkin lymphoma, 339 cases of Hodgkin lymphoma and 2,462 controls. We collected detailed information on occupational history and assessed exposure to meat in general and several types of meat via expert assessment of the questionnaires. The odds ratio (OR) of non-Hodgkin lymphoma for ever occupational exposure to meat was 1.18 (95% confidence interval [CI] 0.95-1.46), that for exposure to beef meat was 1.22 (95% CI 0.90-1.67), and that for exposure to chicken meat was 1.19 (95% CI 0.91-1.55). The ORs were higher among workers with longer duration of exposure. An increased risk among workers exposed to beef meat was mainly apparent for diffuse large B-cell lymphoma (OR 1.49, 95%CI 0.96-2.33), chronic lymphocytic leukemia (OR 1.35, 95% CI 0.78-2.34) and multiple myeloma (OR 1.40, 95%CI 0.67-2.94). The latter 2 types were also associated with exposure to chicken meat (OR 1.55, 95% CI 1.01-2.37, and OR 2.05, 95%CI 1.14-3.69). Follicular lymphoma and T-cell lymphoma, as well as Hodgkin lymphoma did not show any increase in risk. Occupational exposure to meat does not appear to represent an important risk factor of lymphoma, although an increased risk of specific types of non-Hodgkin lymphoma cannot be excluded. (c) 2007 Wiley-Liss, Inc.